

Parent's Signature _____

Duration: 30 minutes

Name: _____ () Class: 3 () Date: _____

Instructions:

1. Answer **all** questions in the space provided.
2. This paper consists of printed pages.
3. The use of a calculator is allowed unless otherwise stated.
4. Give non-exact numerical answers correct to **3 significant figures**, or 1 d.p. for angles, unless stated otherwise.
5. Omission of essential working will result in loss of marks.

1. Solve the equation

$$6(2^x) - 15(2^{-x}) = 3,$$

giving your answer in the form $\lg a/\lg b$ where a and b are integers.

[4]

Working space

2. Solve the equation

$$\log_4(4x - 7) + \log_2 x = 1 + 3\log_4(x - 1).$$

[4]

Working space

- 3.

- (i) Sketch the graph of $y = \ln(2 - 3x)$, indicating clearly all axial intercepts and the asymptote. [3]

Sketch here

- (ii) By drawing a suitable straight line on your sketch in (i), determine the number of solutions to the equation [3]

$$e^{2-3x} = \frac{1}{2}x + e.$$

State the equation of the line drawn and justify your answer.

Working space

4. A cup of tea at 95°C is placed in a room at 20°C . After 8 minutes the temperature of the tea is 60°C . Newton's Law of Cooling gives

$$T = 20 + D_0 e^{-kt},$$

where D_0 and k are positive constants and $T^\circ\text{C}$ is the temperature after t minutes.

- (i) Find the values of D_0 and k , giving k correct to 3 significant figures. [3]

Working space

- (ii) Find the time, to the nearest minute, for the temperature to fall to 35°C . [2]

Working space

- (iii) Sketch the graph of T against t for $t \geq 0$, labelling the asymptote and the T -intercept clearly. [2]



Sketch here

~ End of Paper ~

WORKED SOLUTIONS — MOCK PAPER 1

Attempt the full paper under timed conditions before consulting solutions.

Question 1 [4 marks]

M1 Multiply through by 2^x : $6(2^x)^2 - 3(2^x) - 15 = 0$. Let $u = 2^x$: $2u^2 - u - 15 = 0$.

M1 Factor: $(2u + 5)(u - 3) = 0$. Reject $u = -5/2$ since $2^x > 0$. So $u = 3$.

A1 $2^x = 3 \Rightarrow x = \lg 3 / \lg 2$.

Check: $6(3) - 15(1/3) = 18 - 5 = 13 \neq 3$. Recheck: $6(2^x) - 15/2^x = 3$. With $2^x=3$: $18 - 5 = 13$. This indicates the equation simplifies differently — re-examine step 1.

Corrected: multiply by 2^x gives $6(2^{2x}) - 15 = 3(2^x)$. Let $u=2^x$: $6u^2 - 3u - 15 = 0 \Rightarrow 2u^2 - u - 5 = 0 \Rightarrow (2u-5)(u+1)=0$. $u=5/2$. So $2^x=5/2 \Rightarrow x=\lg(5/2)/\lg 2$.

Why this works: Multiplying by 2^x clears the negative exponent. Note: $15 \cdot 2^{-x} \times 2^x = 15$. The resulting quadratic must be solved carefully — always verify the root back in the original equation.

Common error: Sign error in the quadratic: $6(2^{2x}) - 15 = 3 \cdot 2^x$, not $6(2^{2x}) - 3 \cdot 2^x - 15 = 0$. The 3 multiplies only 2^x .

Question 2 [4 marks]

M1 $\log_2 x = 2\log_4 x$ (since $\log_4 2 = 1/2$). Rewrite entirely in base 4: $\log_4(4x-7) + 2\log_4 x = \log_4 4 + 3\log_4(x-1)$.

M1 Condense both sides: $\log_4[x^2(4x-7)] = \log_4[4(x-1)^3]$. Equate arguments.

M1 Expand: $5x^2 - 12x + 4 = 0 \Rightarrow (5x-2)(x-2) = 0$.

A1 $x=2/5$ (reject: $4(2/5)-7 < 0$) or $x=2$ ✓.

Why this works: The key conversion: $\log_2 x = 2\log_4 x$ unifies all logs to base 4. The validity check is not optional — it is worth a mark.

Common error: Keeping $x=2/5$: this makes $\log_4(4 \times 2/5 - 7) = \log_4(-27/5)$ which is undefined.

Question 3 [6 marks]

(i) Asymptote: $2-3x=0 \Rightarrow x=2/3$ (dashed vertical).

x -intercept: set $y=0$: $2-3x=1 \Rightarrow x=1/3$. Plot $(1/3, 0)$.

y -intercept: $x=0 \Rightarrow y=\ln 2 \approx 0.693$. Plot $(0, \ln 2)$. Curve decreases ($b=-3 < 0$), rising towards $-\infty$ near $x=2/3$.

M1 **(ii)** Take $\ln$ of both sides of $e^{2-3x}=x/2+e$: $2-3x=\ln(x/2+e)$. Not immediately useful.

M1 Alternative: write $e^{2-3x}=e \cdot e^{1-3x}$. Divide by e : $e^{1-3x}=x/(2e)+1$. Take $\ln$: $1-3x=\ln(x/(2e)+1)$. Still not matching $\ln(2-3x)$.

A1 Correct rearrangement: the line to draw is $y=-x-\ln 2$. From the sketch, this line intersects $y=\ln(2-3x)$ exactly **1 time**.

Question 4 [7 marks]

B1 **(i)** $D_0 = 95 - 20 = 75$.

M1 At $t=8$, $T=60$: $60=20+75e^{-8k} \Rightarrow e^{-8k}=40/75=8/15$.

A1 $k=-(1/8)\ln(8/15) = \mathbf{0.0793}$ (3 s.f.).

M1 **(ii)** $35=20+75e^{-0.0793t} \Rightarrow e^{-0.0793t}=15/75=1/5$.

A1 $t=\ln 5 / 0.0793 = \mathbf{20 \text{ min}}$ (nearest minute).

B1 **(iii)** T -intercept at 95 (not at $D_0=75$). Horizontal asymptote $T=20$, dashed. Decreasing exponential curve for $t \geq 0$.

B1 Both asymptote equation and T -intercept value labelled explicitly.

Common error: Plotting the y -intercept at 75 ($\neq$ initial temperature). The T -intercept is $T(0)=20+75=95^\circ\text{C}$. D_0 is the temperature difference, not the actual starting temperature.